Theory Section Draft

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0.1 An introduction to gravitational lensing

Gravitational lensing is one of the most famous predictions of Einstein's theory of relativity. Massive objects bend spacetime, and light follows the curvature of spacetime, meaning that light will follow a curved path around a massive object. For a light ray passing near a mass M, general relativity predicts a deflection angle

$$\alpha = \frac{4GM}{c^2 b} = \frac{2r_g}{b} \tag{1}$$

where G is the gravitational constant, c is the speed of light, r_b is the gravitational radius and b is a quantity known as the impact parameter, the shortest distance between the path of the light ray and the lensing mass. The geometry of a lensing event is shown below in Figure 1

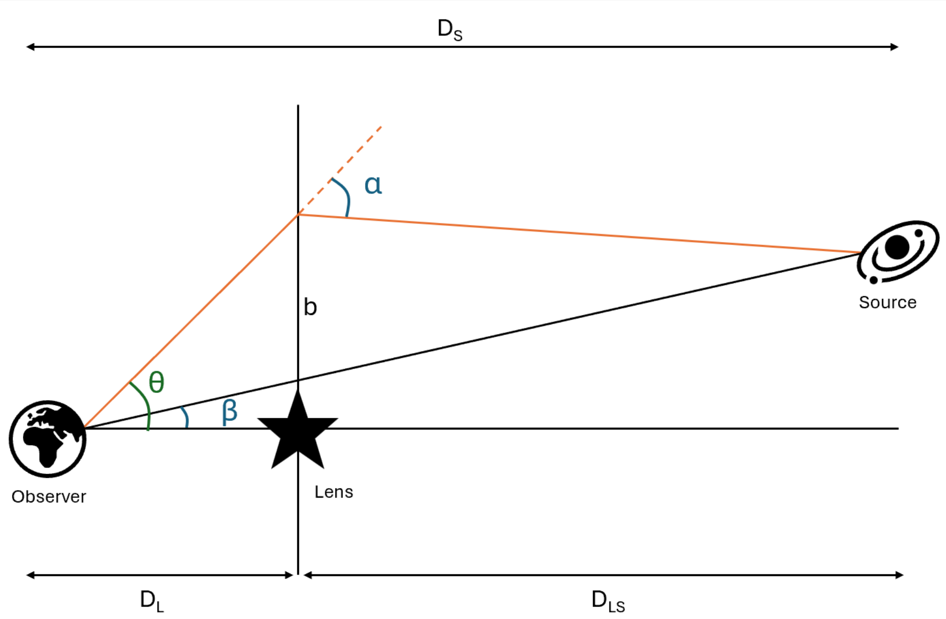

Figure 1: Basic lensing geometry

Where we have assumed that all gravitational deflection of light occurs in a single plane, that is, the thickness of the lensing mass is tiny compared to the distances between observer, lens and source. Lensing events are often categorised into 3 types: strong, weak and microlensing.

- **Strong lensing**: Involves massive lenses (such as galaxies) and we observe large, obvious distortions of the source such as rings or multiple images.
- **Weak lensing**: Still involves massive lenses but with less favourable alignment, so images appear sheared. Often these shears can be analysed statistically across many background galaxies if they can't be identified directly.
- **Microlensing**: The lens has a smaller mass, such as stars, planets and brown dwarfs. The images are too small to be resolved from each other. Instead we observe a temporary brightening of the source.

The lens equation relates the true position of the background source to the apparent position of the lensed image, given by

$$\beta = \theta - \frac{D_{LS}}{D_S}\alpha\left(\theta\right) \tag{2}$$

In the case where the source, lens and observer all lie along a straight line, the lens equation predicts a circular image called an Einstein ring, with the 'Einstein radius' being the radius of this ring. Otherwise, we get 2 images. When an object is lensed, the image appears magnified. The image covers a larger angular area and each bit of that area has the same surface brightness as before, so the total flux increases. It might seem like this violates energy conservation or that the lens creates light, but this is not the case. The lens bends light rays towards us that would have missed us, so we receive more of the light that was already being emitted. The magnification from a point source is given by

$$\mu(u) = \frac{u^2 + 2}{u\sqrt{(u^2 + 4)}} \tag{3}$$

where u is the angular separation between lens and source in terms of the Einstein radius.

0.2 The sun as a gravitational lens

The deflection angle that a light ray will be bent as it passes the sun depend on the impact parameter. This means that rays passing closer to the sun will be bent more than those passing further away. For a light ray passing the sun at an angle α, it will intersect the optical axis at

$$z(b) = \frac{b^2}{2r_g} \tag{4}$$

this means that the solar gravitational lens doesn't focus rays into a point, but a line. Using a wave optics analysis, we can find the maximum on-axis amplification, given by

$$\mu_{max} = 4\pi^2 \frac{r_g}{\lambda} \tag{5}$$

depending on the wavelength that we use, we are looking at an amplification of around 10^{10}. If we say the gain $G \approx \mu$, then in decibels the gain is about 100dB. The sun's corona is a plasma which conatains many free electrons, so it interacts with electromagnetic waves in a frequency-dependent way. It acts like a refractive medium, bending rays away from the sun, the opposite effect to gravitational lensing. The corona will reduce the peak amplification of the solar gravitational lens and will make the point spread function to become wider and less intense.